Constantinos Paschalis

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Portfolio

Profile

MSc Energy Science and Technology student at ETH Zürich specializing in energy systems modeling and optimization. Experience in techno-economic analysis, heat pump systems, and CO_2 transport modeling across academic and industry settings.

Education

ETH Zürich, Zurich, Switzerland — 2024 – Present
MSc Energy Science and Technology

Texas A&M University, Doha, Qatar — 2024
Energy and Sustainability Summer School

University of Cyprus, Nicosia, Cyprus — 2020 – 2024
BSc Mechanical Engineering

Experience

EnergyIntel Services Ltd, Nicosia, Cyprus — Jun 2025 – Aug 2025
R&D Intern — Modeled thermal energy storage systems and evaluated fluid selection using multi-criteria analysis.

Texas A&M University, College Station, TX, USA — Jun 2023 – Jul 2023
Research Assistant (HVAC) — Performed hardware-in-the-loop and simulation-based analysis of HVAC systems under varying climate conditions.

University of Cyprus, Nicosia, Cyprus — Nov 2022 – Nov 2022
Department & School Representative — Represented students in faculty meetings, contributing to academic policy discussions and coordinating communication between students and administration.

Projects

CO_2 Transport Network Resilience – ETH Zürich
Built techno-economic model of CO_2 transport resilience, showing resilience is cost-effective only under high utilization and prolonged failures, with temporary storage as the optimal strategy.

Mega Heat Pump Economic Evaluation – ETH Zürich & Everllence
Developed techno-economic model integrating thermodynamics and electricity markets, demonstrating revenue potential from FCR participation under varying market conditions.

Household Energy Optimization – ETH Zürich
Built MILP model and local app, enabling optimal appliance selection under dynamic pricing and CO_2 constraints.

Skills

Programming: Python, MATLAB
Modeling: ANSYS, SolidWorks
Core: Energy Systems, Thermodynamics, Optimization
Languages: Greek (Native), English (Fluent)

Awards

ETEK, AHK, Elysse Irrigation LTD (2024) — Top Mechanical Engineering Graduate (9.06/10)
MUSKITA Awards (2022, 2023) — Highest academic performance (2nd & 3rd year)